A 16-Vertex Counterexample to Path Minimality for Laplacian Quantum-Walk Laziness

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Abstract

For a continuous-time quantum walk generated by the combinatorial Laplacian L_G of a finite simple graph, let $\widehat{M}(G)$ be the infinite-time Cesàro average of the transition-probability matrix and set $\tau(G) = \operatorname{Tr}\widehat{M}(G)$. It was asked whether the path P_n minimizes τ among all graphs on n vertices. We give an exact counterexample. On $\mathbb{Z}/16\mathbb{Z}$, start with the circulant having cyclic differences $\{\pm 2, \pm 3, \pm 4, \pm 6\}$ and delete the edges $\{0, 13\}$ and $\{5, 8\}$. The resulting connected, unweighted, loopless simple graph G has 62 edges and satisfies

$$\tau(G) = \frac{1673}{1192} < \frac{23}{16} = \tau(P_{16}), \qquad \tau(G) - \tau(P_{16}) = -\frac{81}{2384}.$$

The sign is certified in exact rational arithmetic by orthogonal projection onto the commutant of L_G; a structurally independent full-eigenspace calculation provides a numerical cross-check. Thus the path is not a global minimizer of long-time return over all unweighted graph topologies.

Keywords: continuous-time quantum walk; average mixing matrix; graph Laplacian; algebraic graph theory; exact counterexample

1 Introduction

Let G be a finite graph with combinatorial Laplacian $L_G = D_G - A_G$. The continuous-time walk generated by L_G has transition matrix $U_G(t) = \exp(-itL_G)$. Entrywise squared amplitudes generally do not converge, but their infinite-time Cesàro average does. This average mixing matrix is a spectral graph invariant introduced and developed in the study of continuous quantum walks [2]. Its trace measures the average long-time tendency to return to the initial vertex; recent work calls this trace the *laziness* of the walk [4].

Godsil, Guo and Sobchuk studied extremal diagonal behavior and tabulated Laplacian trace minima through eight vertices [3]. Mohan, Tamon, Xu and Zhan subsequently developed further extremal results and, after reporting data through nine vertices, asked whether P_n is among the minimizers over all n-vertex graphs [4, Section 8(ii)]. The motivating formula is Godsil's exact average mixing matrix for the Laplacian walk on a path [2, Theorem 5.3].

We answer that universal question in the negative with one connected graph on 16 vertices. The witness is specified both by a concise cyclic construction and by a complete edge list. Its strict deficit is an exact rational number, so neither a finite observation time nor a floating-point tolerance determines the conclusion. The result concerns only this counterexample: it does not classify the minimizers at order 16, identify the smallest order of failure, or give an infinite family or replacement bound.

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2 Laplacian walk and long-time return

Throughout, $G = (V, E)$ is finite, undirected, unweighted, loopless and simple, with $|V| = n$. Disconnected graphs are allowed in the universal question, although the witness below is connected. On the single-excitation Hilbert space $\mathbb{C}^V$, take

$$H = L_G = D_G - A_G, \qquad U_G(t) = e^{-itL_G}, \qquad \hbar = 1,$$

with unit edge coupling. There is no additional potential or continuum boundary. All quantities below are exact finite-dimensional quantities; no perturbation series, thermodynamic limit or time discretization is used.

Let $\Lambda(G)$ be the set of distinct eigenvalues of L_G, and let E_λ denote the orthogonal projector onto the full λ-eigenspace. The spectral decomposition gives

$$U_G(t) = \sum_{\lambda \in \Lambda(G)} e^{-it\lambda} E_\lambda.$$

Time averaging removes terms with unequal frequencies, hence

$$\widehat{M}(G)_{uv} = \lim_{T\to\infty} \frac{1}{T} \int_0^T |U_G(t)_{uv}|^2 \, dt = \sum_{\lambda \in \Lambda(G)} (E_\lambda)^2_{uv}. \tag{1}$$

The full eigenspace projector is essential at a degenerate eigenvalue. We define

$$\tau(G) = \operatorname{Tr} \widehat{M}(G) = \sum_{v \in V} \sum_{\lambda \in \Lambda(G)} (E_\lambda)^2_{vv}. \tag{2}$$

Consequently, $\tau(G)/n$ is the long-time return probability averaged over a uniformly selected initial vertex. Sources that use e^{+itL_G} give the same observable: because L_G is real symmetric, changing the sign of time complex-conjugates $U_G(t)$ and leaves all probabilities unchanged.

For P_n, let J be the all-ones matrix and let R reverse the vertex order. The published formula is

$$\widehat{M}(P_n) = \frac{(n-1)J + (n/2)(I + R)}{n^2}. \tag{3}$$

When n is even, R has no fixed vertex, and tracing (3) gives

$$\tau(P_n) = \frac{3}{2} - \frac{1}{n}, \qquad \tau(P_{16}) = \frac{23}{16}. \tag{4}$$

3 The 16-vertex witness

Write the vertices as $\mathbb{Z}/16\mathbb{Z}$. Begin with the undirected circulant

$$E_0 = \{\{x, x+s\} : x \in \mathbb{Z}/16\mathbb{Z},\ s \in \{2, 3, 4, 6\}\}, \tag{5}$$

where addition is modulo 16, and define

$$E(G) = E_0 \setminus \{\{0, 13\}, \{5, 8\}\}. \tag{6}$$

The two deleted step-3 edges are exchanged by the half-turn $x \mapsto x + 8$; their endpoints are not individually antipodal. The base circulant is 8-regular with 64 edges. The deleted edges are disjoint,

so G has 62 edges and degree sequence $(8^{12}, 7^4)$, with degree-seven vertices $0, 5, 8, 13$. It contains the spanning path

$$0, 2, 4, 6, 8, 10, 12, 14, 1, 3, 5, 7, 9, 11, 13, 15,$$

which proves connectivity. The construction uses only unit, undirected edges and introduces neither loops nor multiple edges.

Figure 1 displays the graph construction and the exact return comparison. For complete independent transcription, Table 1 lists every edge once: the row indexed by u gives precisely the neighbors $v > u$.

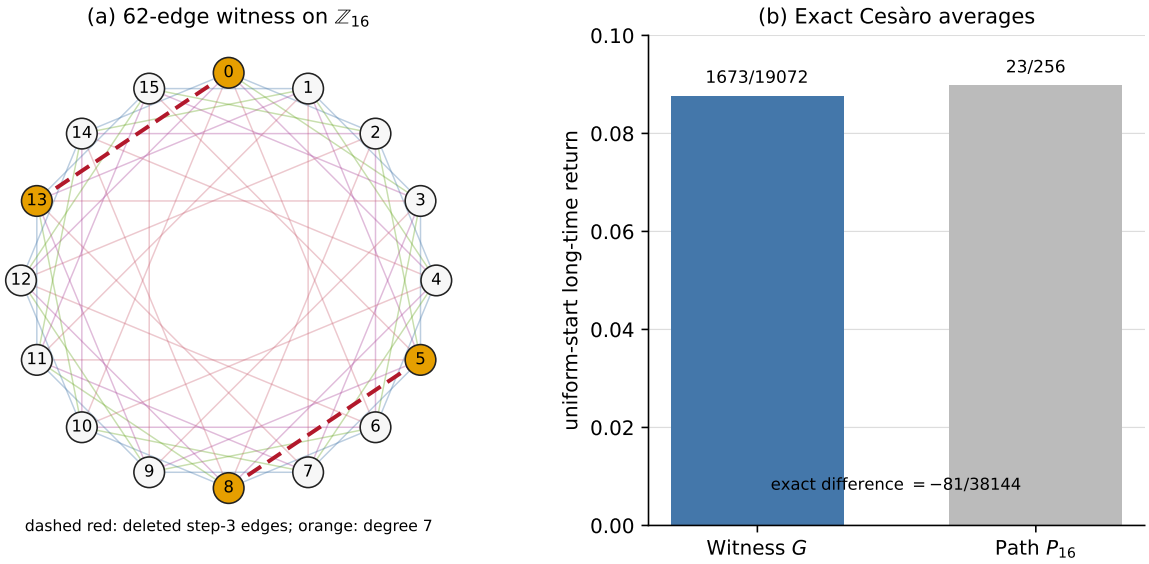


Figure 1: (a) The witness graph, with the two deleted step-3 edges shown as red dashes and the four degree-seven vertices in orange. Present edges are colored by cyclic step. (b) Uniform-start long-time return probabilities, plotted from zero and labeled by their exact values.

Table 1: Complete zero-based edge list for G, grouped by the smaller endpoint. Empty terminal rows for 14 and 15 are omitted.

u	all $v > u$ with $\{u, v\} \in E(G)$	u	all $v > u$ with $\{u, v\} \in E(G)$
0	2, 3, 4, 6, 10, 12, 14	7	9, 10, 11, 13
1	3, 4, 5, 7, 11, 13, 14, 15	8	10, 11, 12, 14
2	4, 5, 6, 8, 12, 14, 15	9	11, 12, 13, 15
3	5, 6, 7, 9, 13, 15	10	12, 13, 14
4	6, 7, 8, 10, 14	11	13, 14, 15
5	7, 9, 11, 15	12	14, 15
6	8, 9, 10, 12	13	15

4 Exact commutant certificate

We use the commutant representation of average mixing [1]. The following formulation makes the exact calculation explicit. Matrices are vectorized by stacking columns, and $D_v = e_v e_v^{\mathsf{T}}$ and $d_v = \mathrm{vec}(D_v)$.

Lemma 1. *Let L be a real symmetric $n \times n$ matrix. If the columns of B form a basis for the vectorized commutant*

$$\mathcal{C}(L) = \{X \in \mathbb{R}^{n\times n} : LX = XL\},$$

then

$$Q = B(B^{\mathsf{T}}B)^{-1}B^{\mathsf{T}} \tag{7}$$

is its Frobenius-orthogonal projector and

$$\widehat{M}(L)_{uv} = d_u^{\mathsf{T}} Q d_v, \qquad \tau(L) = \sum_{v=0}^{n-1} d_v^{\mathsf{T}} Q d_v. \tag{8}$$

Proof. If $L = \sum_\lambda \lambda E_\lambda$, the Frobenius projection onto $\mathcal{C}(L)$ is $X \mapsto \sum_\lambda E_\lambda X E_\lambda$. Consequently,

$$\langle D_u, \sum_\lambda E_\lambda D_v E_\lambda \rangle_F = \sum_\lambda (E_\lambda)_{uv}(E_\lambda)_{vu} = \widehat{M}(L)_{uv}.$$

Equation (7) is the standard matrix of the orthogonal projection onto the column space of a full-rank B. Restricting its rows and columns to the diagonal matrix units proves (8). □

For the Laplacian of the edge list in Table 1, exact integer arithmetic gives the characteristic polynomial

$$\begin{aligned}\chi_L(x) = {} & x(x-10)(x-8)(x-6)(x^2-20x+98)(x^2-16x+62)\\ & \times (x^4-32x^3+368x^2-1768x+2874)\\ & \times (x^4-32x^3+380x^2-1984x+3842).\end{aligned} \tag{9}$$

The exact check $\gcd(\chi_L, \chi_L') = 1$ shows that the spectrum is simple. Thus the minimal polynomial has degree 16 and the commutant has the basis

$$B = \left[\mathrm{vec}(I)\ \ \mathrm{vec}(L)\ \ \cdots\ \ \mathrm{vec}(L^{15})\right]. \tag{10}$$

Equivalently, if

$$C = I \otimes L - L^{\mathsf{T}} \otimes I,$$

then $CB = 0$, the 16 columns of B are independent, and they span $\ker C$.

The evaluation can be compressed to a 16×16 rational calculation. Let

$$\Gamma = B^{\mathsf{T}}B, \qquad R_{v,k} = (L^k)_{vv}, \quad 0 \le v, k \le 15. \tag{11}$$

Selecting the diagonal rows of B in Lemma 1 gives

$$\widehat{M}(G) = R\Gamma^{-1}R^{\mathsf{T}}, \qquad \tau(G) = \mathrm{Tr}\left(R\Gamma^{-1}R^{\mathsf{T}}\right). \tag{12}$$

Every entry of L, R and Γ is an integer. The accompanying exact script verifies $CB = 0$ and $\Gamma\Gamma^{-1} = I$ symbolically before applying (12); no eigenvalue approximation enters this route.

Proposition 1. *For the graph G of* (5)–(6)*,*

$$\tau(G) = \frac{1673}{1192}.$$

Proof. Constructing the integer Laplacian from Table 1, forming the power basis (10), and evaluating (12) in $\mathbb{Q}$ gives

$$\mathrm{Tr}(R\Gamma^{-1}R^{\mathsf{T}}) = \frac{1673}{1192}.$$

The input edge table, factorization (9), exact matrix guards and rational output are included in the machine-readable supplement, together with the short reconstruction program. □

A separate calculation constructs the edges from the cyclic rule rather than from Table 1, diagonalizes the resulting symmetric Laplacian, groups full eigenspace projectors at two gap tolerances, and checks residuals, orthogonality, projector completeness and idempotence. Both tolerances find 16 distinct spectral groups and give 1.4035234899328854, differing from 1673/1192 by 4.44×10^{-16}. This floating-point route is a cross-check only; the strict inequality below rests on the rational calculation.

5 Disproof and physical interpretation

Combining the proposition with (4) gives the exact identity

$$\tau(G) - \tau(P_{16}) = \frac{1673}{1192} - \frac{23}{16} = -\frac{81}{2384} < 0. \tag{13}$$

Theorem 1. *The statement $\tau(X) \geq \tau(P_n)$ for every $n \geq 2$ and every undirected, unweighted, loopless simple n-vertex graph X is false.*

Proof. The graph G in Section 3 belongs to the quantified class, and (13) violates the asserted inequality at $n = 16$. □

For a uniformly chosen initial vertex, division by 16 gives

$$\frac{\tau(G)}{16} = \frac{1673}{19072}, \qquad \frac{\tau(P_{16})}{16} = \frac{23}{256}, \qquad \frac{\tau(G) - \tau(P_{16})}{16} = -\frac{81}{38144}.$$

Thus this unweighted topology has a strictly smaller long-time return than the path in the same finite, closed single-excitation Laplacian model. The claim is not that the propagator converges pointwise, nor that a finite-time experiment has been performed; it is an exact statement about its Cesàro average.

6 Scope and outlook

One strict admissible witness is enough to settle the universal path-minimization question, but it leaves the revised extremal problem open. In particular, the calculation does not establish the minimum at $n = 16$, the smallest order at which a counterexample exists, an equality classification, an infinite counterexample family, or a new all-order lower bound. The available literature comparison is also bounded, especially because the motivating question is recent; priority should therefore be rechecked before submission. A natural next mathematical task is to determine whether the near-circulant mechanism in (6) extends to a controlled family or instead produces an isolated finite-order exception.

Data and code availability

The local supplementary package contains the complete edge data, the exact rational commutant calculation, the independent grouped-eigenspace cross-check, the comparison script, the generated figure source and explicit regeneration instructions. The exact certificate requires only integer and rational linear algebra. No downloaded third-party paper is included in the supplement.

AI-assistance disclosure

Generative AI assisted literature triage, code generation, exact-calculation workflows, verification and manuscript drafting. The named human author must independently check all claims, calculations, citations and wording, approve the authorship metadata, and choose final disclosure language before any submission. This local draft does not represent completed human review.

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